

Virtualized Hardware-Style Next Generation Firewall with AI-Assisted Intrusion Detection and DPI

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Abstract:

With the exponential growth of cyber threats, traditional firewalls that rely solely on IP- and port-based filtering are increasingly inadequate for protecting modern, dynamic network environments. This project proposes the development of a **virtualized Next Generation Firewall (NGFW) appliance** that runs as a dedicated security **gateway VM** with multiple network interfaces (WAN/LAN) so that all client traffic is routed through it—providing a hardware-like deployment experience using commodity virtualization platforms. The appliance integrates **core firewall features** (IP/port filtering, protocol-aware rules, and stateful packet inspection) to enforce baseline network policy. Security depth is enhanced through **Deep Packet Inspection (DPI)** of application-layer protocols such as HTTP, DNS, and FTP to detect malicious patterns, blocked domains, and file transfer activity. An **SSL/TLS handling module** identifies encrypted HTTPS flows, extracts certificate metadata (issuer, validity, CN), and optionally supports proxy-based decryption for controlled content inspection in lab environments. For proactive defense, an **Intrusion Detection and Prevention (IDS/IPS)** layer combines signature/rule-based detection (e.g., SYN flood recognition) with **machine learning models** (Random Forest, SVM) for real-time anomaly detection. A **Threat Intelligence Integration** service ingests public IP/domain blocklists to automatically enforce dynamic reputation-based controls. Operational visibility and control are provided through an **integrated web-based management GUI** hosted inside the firewall VM, enabling administrators to visualize traffic, manage rules, review alerts, examine SSL certificate details, and generate security reports. By packaging advanced NGFW capabilities into a self-contained **virtual firewall appliance with an embedded management interface**, this project delivers a scalable, low-cost, and academically demonstrable platform for modern network security research, learning, and controlled deployment.

The proposed system provides multi-layered protection through the following modules:

1. **Core Firewall Features:** Implements IP and port filtering, protocol-based rules, and stateful packet inspection to manage traffic effectively.
2. **Deep Packet Inspection (DPI):** Analyzes application-layer traffic such as HTTP, DNS, and FTP to detect malicious patterns, blocked domains, and file transfers.
3. **SSL/TLS Handling:** Identifies encrypted HTTPS traffic, extracts SSL certificate details (issuer, validity), and optionally performs content inspection using proxy-based decryption.
4. **Intrusion Detection & Prevention (IDS/IPS):** Combines rule-based detection (e.g., SYN flood attacks) with **Machine Learning models** (Random Forest, SVM) to identify anomalous behavior in real-time.
5. **Threat Intelligence Integration:** Incorporates public threat feeds for blocking blacklisted IP addresses and domains.
6. **Graphical User Interface (GUI):** Provides a user-friendly dashboard to visualize traffic statistics, manage firewall rules, and generate security reports.
7. **Reporting & Logging:** Maintains detailed logs in a database, supports CSV exports, and generates analytics charts for network activity and security events.

The architecture is built on **Python**, utilizing libraries such as **Scapy** for packet capture, **PyShark** for traffic analysis, **Scikit-learn** for ML-based anomaly detection, and **Flask/Django** for the web-based dashboard. The solution will run on a standard Linux environment and leverage open-source tools for SSL handling and DPI. By integrating **DPI, SSL inspection, and AI-based IDS**, this NGFW prototype addresses limitations of conventional firewalls and demonstrates a scalable, intelligent security solution for modern networks.